

Solutions to Dummit & Foote's Abstract Algebra 3rd Edition

JR

October 27th, 2025

Contents

Preface	2
Contributors	3
0 Preliminaries	4
0.1 Basics	4
0.2 Properties of the Integers	6
0.3 $\mathbb{Z}/n\mathbb{Z}$: The Integers Modulo n	9

Preface

This collection contains solutions to the exercises in David S. Dummit and Richard M. Foote’s *Abstract Algebra*, Third Edition—a comprehensive graduate-level text that has become a standard reference in the field. I have worked through these problems over time to deepen my understanding of the material and to create a resource for others studying abstract algebra. While these solutions are written with care and rigor, they are not guaranteed to be error-free; if you discover mistakes or have suggestions for improvement, I welcome corrections.

Solutions are organized by chapter and section, mirroring the structure of the textbook. Within each section, exercises are presented in the order they appear in the book. Each solution aims to be self-contained but frequently references prior results; when a proof depends heavily on another exercise or theorem, I note this explicitly.

Exercise 0.0.1

Most problems in the book are presented in the standard exercise environment. These exercises are typically routine and don’t require advanced mathematical insight beyond the techniques and concepts introduced in the text.

Exercise 0.0.2

Problems in the “special exercise” environment are included because they are particularly important, instructive, or deep. Some are challenging in their proof techniques or conceptual demands; others are simple but illustrate key ideas fundamental to the overall text. These exercises deepen understanding and often provide insight into material developed later in the book.

This is an ongoing project. As I work through the text, I refine earlier solutions and add new ones. While I strive for mathematical correctness, errors may remain—particularly in computational problems or long proofs. I encourage you to read critically, verify claims, and develop your own solutions. The goal is not to provide answers to be memorized, but to illustrate good problem-solving techniques and deepen your understanding of abstract algebra.

I am completing this project independently at my own pace. That said, I genuinely welcome corrections, suggestions, and alternative solutions from the community. If you find errors, have ideas for improvement, or wish to contribute, please open an issue or submit a pull request on GitHub. You may also reach out directly via Reddit or Discord with feedback or discussion. I appreciate the interest in helping this project grow and remain open to community involvement.

My deepest thanks to David S. Dummit and Richard M. Foote for writing *Abstract Algebra*. Their exposition is remarkably clear, their choice of exercises is thoughtful and comprehensive, and their book has been instrumental in my mathematical development. Many of these solutions exist because of the beauty and depth of their presentation.

Finally, I’d like to take a moment and thank the broader mathematics community for maintaining a culture of shared knowledge. The practice of working through textbooks and sharing solutions has a long and valuable tradition, and I am grateful to everyone who contributes to this collective effort. I hope that this project, in its small way, adds to that tradition and proves useful to others in their study of abstract algebra. Please enjoy!

Contributors

As of now, I am independently forming solutions to each exercise myself. However, I welcome any corrections, suggestions, or slicker solutions from others. A (growing) list of contributors will be maintained here. If you'd like to contact me, please reach out on Reddit at [my profile](#) or via Discord, where my username is blanketism. Thank you to everyone who decided to take time out of their busy lives to contribute to this project. I appreciate your help in making it better for everyone.

- greysonbrowser
- coolempire93

0 Preliminaries

0.1 Basics

For Exercise 0.1.1 through Exercise 0.1.4, let

$$M = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

and $\mathcal{B} = \{X \in \mathcal{A} \mid MX = XM\}$, where $\mathcal{A}$ denotes the set of 2×2 matrices with real entries.

Exercise 0.1.1

Determine which of the following elements of $\mathcal{A}$ lie in $\mathcal{B}$:

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Solution. Let M_i denote the i -th matrix in the list. M_1, M_3, M_5, M_6 is M , the zero matrix, I_2 , and the exchange matrix respectively. Among these, M_1, M_3, M_5 belong to $\mathcal{B}$, while $M_6 \notin \mathcal{B}$. Direct calculation shows M_2 and M_4 are not in $\mathcal{B}$. ■

Exercise 0.1.2

Prove that if $P, Q \in \mathcal{B}$, then $P + Q \in \mathcal{B}$ where $+$ denotes the usual sum of two matrices.

Solution. If $P, Q \in \mathcal{B}$, then $M(P + Q) = MP + MQ = PM + QM = (P + Q)M$, so $P + Q \in \mathcal{B}$. ■

Exercise 0.1.3

Prove that if $P, Q \in \mathcal{B}$, then $P \cdot Q \in \mathcal{B}$ where $\cdot$ denotes the usual product of two matrices.

Solution. If $P, Q \in \mathcal{B}$, then $M(PQ) = (MP)Q = (PM)Q = P(MQ) = P(QM) = (PQ)M$, so $PQ \in \mathcal{B}$. ■

Exercise 0.1.4

Find conditions on p, q, r, s which determine precisely when

$$\begin{pmatrix} p & q \\ r & s \end{pmatrix} \in \mathcal{B}.$$

Solution. Let X be the matrix above. Computing $MX = XM$ gives the following system of equations:

$$\begin{cases} p + r = p \\ q + s = p + q \\ r = r \\ s = r + s \end{cases}$$

Simplifying results in $r = 0$ and $p = s$. Then

$$\mathcal{B} = \left\{ \begin{pmatrix} p & q \\ 0 & p \end{pmatrix} \mid p, q \in \mathbb{R} \right\}.$$

Exercise 0.1.5

Determine whether the following functions f are well-defined:

- (a) $f : \mathbb{Q} \rightarrow \mathbb{Z}$ defined by $f(a/b) = a$.
 (b) $f : \mathbb{Q} \rightarrow \mathbb{Q}$ defined by $f(a/b) = a^2/b^2$.

Solution.

- (a) No: $1/2 = 2/4$, but $f(1/2) = 1 \neq 2 = f(2/4)$.
 (b) Yes: $a/b = c/d$ implies $a^2/b^2 = c^2/d^2$, so $f(a/b) = f(c/d)$. ■

Exercise 0.1.6

Determine whether the function $f : \mathbb{R}^+ \rightarrow \mathbb{Z}$ defined by mapping a real number r to the first digit to the right of the decimal point in a decimal expansion of r is well-defined.

Solution. No. Note that $1 = 1.000\dots = 0.999\dots$, but $f(1.000\dots) = 1$ and $f(0.999\dots) = 9$. ■

Exercise 0.1.7

Let $f : A \rightarrow B$ be a surjective map of sets. Prove that the relation

$$a \sim b \iff f(a) = f(b)$$

is an equivalence relation whose equivalence classes are the fibers of f .

Solution. That $\sim$ is an equivalence relation follows from $=$ being an equivalence relation.

The equivalence class of $a \in A$ is $\{x \in A \mid x \sim a\}$. By definition, this is $\{x \in A \mid f(x) = f(a)\}$, which is precisely the fiber of f over $f(a)$. ■

0.2 Properties of the Integers

Exercise 0.2.1

For each of the following pairs of integers a and b , determine their greatest common divisor, their least common multiple, and write their greatest common divisor in the form $ax + by$ for some integers x and y .

- (a) $a = 20, b = 13$
- (b) $a = 69, b = 372$
- (c) $a = 792, b = 275$
- (d) $a = 11391, b = 5673$
- (e) $a = 1761, b = 1567$
- (f) $a = 507885, b = 60808$

Solution.

	(a, b)	$\text{lcm}(a, b)$	$ax + by$
(a)	1	260	$2(20) - 3(13)$
(b)	3	8556	$27(69) - 5(372)$
(c)	11	19800	$8(792) - 23(275)$
(d)	3	21540381	$253(5673) - 126(11391)$
(e)	1	2759487	$-105(1761) + 118(1567)$
(f)	691	44693880	$142(60808) - 17(507885)$

Here, the lcm was computed using $ab = (a, b) \text{lcm}(a, b)$. ■

Exercise 0.2.2

Prove that if the integer k divides the integers a and b , then k divides $as + bt$ for every pair of integers s and t .

Solution. Since $k \mid a$ and $k \mid b$, there exist $x, y \in \mathbb{Z}$ such that $a = kx$ and $b = ky$. Then for any $s, t \in \mathbb{Z}$, we have

$$as + bt = (kx)s + (ky)t = k(xs + yt),$$

which shows that $k \mid (as + bt)$. ■

Exercise 0.2.3

Prove that if n is composite, then there are integers a and b such that n divides ab , but n does not divide either a or b .

Solution. Since n is composite, put $n = ab$ with $1 < a, b < n$. Then $n \mid ab$, but $n \nmid a$ and $n \nmid b$. ■

Exercise 0.2.4

Let a, b and N be fixed integers with a and b nonzero, and let $d = (a, b)$ be the greatest common divisor of a and b . Suppose x_0 and y_0 are particular solutions to $ax + by = N$. Prove, for any integer t , that the integers

$$x = x_0 + \frac{b}{d}t \quad \text{and} \quad y = y_0 - \frac{a}{d}t$$

are also solutions to $ax + by = N$.

REMARK. This is, in fact, the general solution.

Solution.

$$\begin{aligned} ax + by &= a \left(x_0 + \frac{b}{d}t \right) + b \left(y_0 - \frac{a}{d}t \right) \\ &= ax_0 + \frac{ab}{d}t + by_0 - \frac{ab}{d}t \\ &= ax_0 + by_0 = N \end{aligned}$$

■

Exercise 0.2.5

Determine the value $\varphi(n)$ for each integer $n \leq 30$, where φ denotes the Euler φ -function.

Solution.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$\varphi(n)$	1	1	2	2	4	2	6	4	6	4	10	4	12	6	8

n	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
$\varphi(n)$	8	16	6	18	8	12	10	22	8	20	12	18	12	28	8

■

Exercise 0.2.6

Prove the Well Ordering Property of $\mathbb{Z}$ by induction, and prove the minimal element is unique.

Solution. Suppose S has no minimal element, and let T be $\mathbb{Z}^+ \setminus S$, the set of nonnegative integers not in S . Since 0 is minimal in $\mathbb{Z}^+$, then $0 \notin S$, hence $0 \in T$. Fix $k \in \mathbb{Z}^+$, and suppose every j such that $0 \leq j \leq k$ is in T . Observe $k+1 \notin S$, for otherwise it would be minimal in S , since every integer less than or equal to k is in T . Then $k+1 \in T$. By induction, every nonnegative integer is in T , contradicting that S is nonempty. Therefore, S must have a minimal element.

If $s_1, s_2 \in S$ are both minimal, then $s_1 \leq s_2$ and $s_2 \leq s_1$ imply $s_1 = s_2$.

■

Exercise 0.2.7

If p is a prime, prove that there do not exist nonzero integers a and b such that $a^2 = pb^2$.

Solution. Suppose there existed a, b such that $a^2 = pb^2$. If $(a, b) \neq 1$, divide both a and b by (a, b) to get relatively prime integers. Then $p \mid a^2$, which implies $p \mid a$. Put $a = pk$ for $k \in \mathbb{Z}$. Then

$$a^2 = (pk)^2 = p^2k^2 = pb^2,$$

hence $b^2 = pk^2$. But then $p \mid b$, contradicting that $(a, b) = 1$.

■

Exercise 0.2.8

Let p be a prime, $n \in \mathbb{Z}^+$. Find a formula for the largest power of p which divides $n! = n(n-1)(n-2) \cdots 2 \cdot 1$.

HINT. It involves the greatest integer function.

Solution. Note that there exists $k_1 \in \mathbb{Z}^+$ such that $n \geq k_1 p$. Then the integers $p, 2p, \dots, k_1 p$ contribute k_1 factors of p in $n!$. Next, there exists $k_2 \in \mathbb{Z}^+$ such that $n \geq k_2 p^2$. Then the integers $p^2, 2p^2, \dots, k_2 p^2$ contribute k_2 additional factors of p in $n!$. We continue this process until some $k_m \in \mathbb{Z}^+$ such that $n \geq k_m p^m$ but $n < k_{m+1} p^{m+1}$. The largest power of p which divides $n!$ is then given by

$$\sum_{i=1}^m k_i = \sum_{i=1}^m \left\lfloor \frac{n}{p^i} \right\rfloor. \quad \blacksquare$$

Exercise 0.2.9

Write a computer program to determine the greatest common divisor (a, b) of two integers a and b and to express (a, b) in the form $ax + by$ for some integers x and y .

Solution. Please see `euclidean_algorithm.py` for the solution. ■

Exercise 0.2.10

Prove for any given positive integer N there exist only finitely many integers n with $\varphi(n) = N$ where φ denotes Euler's φ -function. Conclude in particular that φ tends to infinity as n tends to infinity.

Solution. Fix $N \in \mathbb{Z}^+$, and consider $n \in \mathbb{Z}^+$ such that $\varphi(n) = N$. Let $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$ be the prime factorization of n . Then

$$\varphi(n) = \varphi(p_1^{\alpha_1}) \varphi(p_2^{\alpha_2}) \cdots \varphi(p_k^{\alpha_k}) = \prod_{i=1}^k p_i^{\alpha_i-1} (p_i - 1).$$

Since $\varphi(n) = N$, there are only finitely many possibilities for the primes p_i and their exponents α_i that satisfy this equation. Hence, there are only finitely many integers n such that $\varphi(n) = N$.

If φ was finite as n tends to infinity, then there would exist some $M \in \mathbb{Z}^+$ such that $\varphi(n) \leq M$ for infinitely many n . However, this contradicts the fact that for each fixed $N \in \mathbb{Z}^+$, there are only finitely many n such that $\varphi(n) = N$. Hence, φ must tend to infinity as n tends to infinity. ■

Exercise 0.2.11

Prove that if d divides n then $\varphi(d)$ divides $\varphi(n)$ where φ denotes Euler's φ -function.

Solution. Let $n \in \mathbb{Z}^+$ with $d \mid n$. Put their prime factorizations as

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}, \quad d = p_1^{\beta_1} p_2^{\beta_2} \cdots p_j^{\beta_j},$$

where $j \leq k$ and $1 \leq \beta_i \leq \alpha_i$ for all $1 \leq i \leq j$. Then $\varphi(d) \mid \varphi(n)$, since

$$\frac{\varphi(n)}{\varphi(d)} = \prod_{i=1}^j \frac{p_i^{\alpha_i-1} (p_i - 1)}{p_i^{\beta_i-1} (p_i - 1)} \cdot \prod_{i=j+1}^k p_i^{\alpha_i-1} (p_i - 1) = \prod_{i=1}^j p_i^{\alpha_i-\beta_i} \cdot \prod_{i=j+1}^k p_i^{\alpha_i-1} (p_i - 1) \in \mathbb{Z}. \quad \blacksquare$$

0.3 $\mathbb{Z}/n\mathbb{Z}$: The Integers Modulo n **Exercise 0.3.1**

Write down explicitly all the elements in the residue classes of $\mathbb{Z}/18\mathbb{Z}$.

Solution. Note that $\bar{x} = \{x + 18k \mid k \in \mathbb{Z}\}$. Explicitly, we have:

$$\begin{array}{lll} \bar{0} = \{0, 18, -18, 36, -36, \dots\} & \bar{1} = \{1, 19, -17, 37, -35, \dots\} & \bar{2} = \{2, 20, -16, 38, -34, \dots\} \\ \bar{3} = \{3, 21, -15, 39, -33, \dots\} & \bar{4} = \{4, 22, -14, 40, -32, \dots\} & \bar{5} = \{5, 23, -13, 41, -31, \dots\} \\ \bar{6} = \{6, 24, -12, 42, -30, \dots\} & \bar{7} = \{7, 25, -11, 43, -29, \dots\} & \bar{8} = \{8, 26, -10, 44, -28, \dots\} \\ \bar{9} = \{9, 27, -9, 45, -27, \dots\} & \bar{10} = \{10, 28, -8, 46, -26, \dots\} & \bar{11} = \{11, 29, -7, 47, -25, \dots\} \\ \bar{12} = \{12, 30, -6, 48, -24, \dots\} & \bar{13} = \{13, 31, -5, 49, -23, \dots\} & \bar{14} = \{14, 32, -4, 50, -22, \dots\} \\ \bar{15} = \{15, 33, -3, 51, -21, \dots\} & \bar{16} = \{16, 34, -2, 52, -20, \dots\} & \bar{17} = \{17, 35, -1, 53, -19, \dots\} \quad \blacksquare \end{array}$$

Exercise 0.3.2

Prove that the distinct equivalence classes in $\mathbb{Z}/n\mathbb{Z}$ are precisely $\bar{0}, \bar{1}, \bar{2}, \dots, \overline{n-1}$.

HINT. Use the Division Algorithm.

Solution. Let $m \in \mathbb{Z}$. By the Division Algorithm, there exists unique $q, r \in \mathbb{Z}$ such that

$$m = nq + r, \quad 0 \leq r < n.$$

Then $m \equiv r \pmod{n}$, so $m \in \bar{r}$. By definition, $\bar{r}$ is one of $\bar{0}, \bar{1}, \dots, \overline{n-1}$. Thus, every equivalence class in $\mathbb{Z}/n\mathbb{Z}$ is one of $\bar{0}, \bar{1}, \dots, \overline{n-1}$. Moreover, they are all distinct. Otherwise, if $\bar{x} = \bar{y}$ for $0 \leq x, y < n$, then $x \equiv y \pmod{n}$, or $n \mid (x - y)$. However, since $-n < x - y < n$, we must have $x - y = 0$, or $x = y$. $\blacksquare$

Exercise 0.3.3

Prove that if

$$a = a_n 10^n + a_{n-1} 10^{n-1} + \dots + a_1 10 + a_0$$

is any positive integer, then

$$a \equiv (a_n + a_{n-1} + \dots + a_1 + a_0) \pmod{9}.$$

HINT. Use the fact that $10 \equiv 1 \pmod{9}$.

REMARK. Note that this is the usual arithmetic rule that the remainder after division by 9 is the same as the sum of the decimal digits mod 9—in particular an integer is divisible by 9 if and only if the sum of its digits is divisible by 9.

Solution. Using the hint, then

$$\begin{aligned} a &\equiv (a_n 10^n + a_{n-1} 10^{n-1} + \dots + a_1 10 + a_0) \pmod{9} \\ &\equiv (a_n 1^n + a_{n-1} 1^{n-1} + \dots + a_1 1 + a_0) \pmod{9} \\ &= (a_n + a_{n-1} + \dots + a_1 + a_0) \pmod{9} \quad \blacksquare \end{aligned}$$

Exercise 0.3.4

Compute the remainder when 37^{100} is divided by 29.

Solution. Modding 37^{100} by 29 results in 23. ■

Exercise 0.3.5

Compute the last two digits of 9^{1500} .

Solution. Modding 9^{1500} by 100 results in 1. ■

Exercise 0.3.6

Prove that the square of the elements in $\mathbb{Z}/4\mathbb{Z}$ are just $\bar{0}$ and $\bar{1}$.

Solution.

$$0^2 = 0 \equiv 0 \pmod{4}$$

$$1^2 = 1 \equiv 1 \pmod{4}$$

$$2^2 = 4 \equiv 0 \pmod{4}$$

$$3^2 = 9 \equiv 1 \pmod{4}$$

■

Exercise 0.3.7

Prove that for any integers a and b that $a^2 + b^2$ never leaves a remainder of 3 when divided by 4.

HINT. Use [Exercise 0.3.6](#).

Solution. By [Exercise 0.3.6](#), a^2 and b^2 are both either 0 mod 4 or 1 mod 4, so their sum can be:

$$0 + 0 \equiv 0 \pmod{4}$$

$$0 + 1 \equiv 1 \pmod{4}$$

$$1 + 0 \equiv 1 \pmod{4}$$

$$1 + 1 \equiv 2 \pmod{4}$$

In any of the sums, there is no remainder of 3 when dividing by 4. ■

Exercise 0.3.8

Prove that the equation $a^2 + b^2 = 3c^2$ has no solutions in nonzero integers a , b and c .

HINT. Consider the equation mod 4 as in the previous two exercises and show that a , b and c would all have to be divisible by 2. Then each of a^2 , b^2 and c^2 has a factor of 4 and by dividing through by 4 show that there would be a smaller set of solutions to the original equation. Iterate to reach a contradiction.

Solution. We consider $a^2 + b^2 = 3c^2$ in mod 4. By [Exercise 0.3.7](#), $a^2 + b^2$ is either 0, 1, or 2 mod 4. By [Exercise 0.3.6](#), c^2 is either 0 or 1 mod 4, so that $3c^2$ is either 0 or 3 mod 4. Thus, both sides must be 0 mod 4.

It is clear that $4 \mid 3c^2$ implies $4 \mid c^2$, and hence $2 \mid c$. Observe that if a and b were both odd, then $a^2 + b^2 \equiv 2 \pmod{4}$. If one of a or b were odd and the other even, then $a^2 + b^2 \equiv 1 \pmod{4}$. Thus, both a and b must be even. Put $a = 2\alpha$, $b = 2\beta$, and $c = 2\gamma$ for nonzero $\alpha, \beta, \gamma \in \mathbb{Z}$. Substituting into the original equation gives

$$(2\alpha)^2 + (2\beta)^2 = 3(2\gamma)^2.$$

We may divide by 4 to obtain $\alpha^2 + \beta^2 = 3\gamma^2$, which is the same form as the original equation but with integers with smaller absolute values. Repeating this process leads to an infinite descent, which is impossible. Therefore, there are no solutions in nonzero integers a, b , and c . ■

Exercise 0.3.9

Prove that the square of any odd integer always leaves a remainder of 1 when divided by 8.

Solution. Put $a = 2k + 1$ for $k \in \mathbb{Z}$. Then $a^2 = 4k(k + 1) + 1 \equiv 1 \pmod{8}$ since $k(k + 1)$ is always even. ■

Exercise 0.3.10

Prove that the number of elements of $(\mathbb{Z}/n\mathbb{Z})^\times$ is $\varphi(n)$, where φ denotes the Euler φ -function.

Solution. It suffices to show that $\bar{a} \in (\mathbb{Z}/n\mathbb{Z})^\times$ if and only if $(a, n) = 1$.

($\Rightarrow$) If $\bar{a} \in (\mathbb{Z}/n\mathbb{Z})^\times$, then there exists $\bar{b} \in \mathbb{Z}/n\mathbb{Z}$ such that $\bar{a} \cdot \bar{b} = \bar{1}$. Then $ab \equiv 1 \pmod{n}$ implies the existence of $k \in \mathbb{Z}$ such that $ab - 1 = kn$, or $ab - kn = 1$. Thus, $(a, n) = 1$.

($\Leftarrow$) By $(a, n) = 1$, there exists $x, y \in \mathbb{Z}$ such that $ax + ny = 1$, or $ax = 1 - ny$. We then have $ax \equiv 1 \pmod{n}$, so that $\bar{a} \cdot \bar{x} = \bar{1}$. Thus, $\bar{a} \in (\mathbb{Z}/n\mathbb{Z})^\times$. ■

Exercise 0.3.11

Prove that if $\bar{a}, \bar{b} \in (\mathbb{Z}/n\mathbb{Z})^\times$, then $\bar{a} \cdot \bar{b} \in (\mathbb{Z}/n\mathbb{Z})^\times$.

Solution. Since $\bar{a}, \bar{b} \in (\mathbb{Z}/n\mathbb{Z})^\times$, there exist $\bar{c}, \bar{d} \in \mathbb{Z}/n\mathbb{Z}$ such that $\bar{a} \cdot \bar{c} = \bar{b} \cdot \bar{d} = \bar{1}$. Then

$$(\bar{a} \cdot \bar{b})(\bar{c} \cdot \bar{d}) = (\bar{a} \cdot \bar{c})(\bar{b} \cdot \bar{d}) = \bar{1} \cdot \bar{1} = \bar{1},$$

so that $\bar{a} \cdot \bar{b} \in (\mathbb{Z}/n\mathbb{Z})^\times$. ■

Exercise 0.3.12

Let $n \in \mathbb{Z}, n > 1$ and let $a \in \mathbb{Z}$ with $1 \leq a \leq n$. Prove if a and n are not relatively prime, there exists an integer b with $1 \leq b < n$ such that $ab \equiv 0 \pmod{n}$ and deduce that there cannot be an integer c such that $ac \equiv 1 \pmod{n}$.

Solution. Let $(a, n) = d > 1$. Then $a/d, n/d \in \mathbb{Z}$ and $1 \leq a/d < n/d$. Put $b = n/d$, so that $1 \leq b < n$ and

$$ab = a(n/d) = (a/d)n \equiv 0 \pmod{n}.$$

If there existed some $c \in \mathbb{Z}$ such that $ac \equiv 1 \pmod{n}$, then $(ac)b \equiv b \pmod{n}$. However, $ab \equiv 0 \pmod{n}$, so that $0 \equiv b \pmod{n}$, which is impossible since $1 \leq b < n$. ■

Exercise 0.3.13

Let $n \in \mathbb{Z}, n > 1$ and let $a \in \mathbb{Z}$ with $1 \leq a \leq n$. Prove that if a and n are relatively prime then there is an integer c such that $ac \equiv 1 \pmod{n}$.

HINT. Use the fact that the GCD of two integers is a $\mathbb{Z}$ -linear combination of the integers.

Solution. Since $(a, n) = 1$, there exists $c, x \in \mathbb{Z}$ such that $ac + nx = 1$. Then $ac \equiv 1 \pmod{n}$. ■

Exercise 0.3.14

Conclude from the previous two exercises that $(\mathbb{Z}/n\mathbb{Z})^\times$ is the set of elements $\bar{a}$ of $\mathbb{Z}/n\mathbb{Z}$ with $(a, n) = 1$ and hence prove Proposition 4. Verify this directly in the case $n = 12$.

Solution. The previous two exercises show that a and n are relatively prime if and only if there exists b such that $ab \equiv 1 \pmod{n}$, which is exactly the proposition. For $n = 12$, the elements 1, 5, 7, 11 are relatively prime to 12, so that $(\mathbb{Z}/12\mathbb{Z})^\times = \{\bar{1}, \bar{5}, \bar{7}, \bar{11}\}$ whose inverses are $\bar{1}, \bar{5}, \bar{5}, \bar{11}$ respectively. ■

Exercise 0.3.15

For each of the following pairs of integers a and n , show that a is relatively prime to n and determine the multiplicative inverse of $\bar{a}$ in $\mathbb{Z}/n\mathbb{Z}$.

- (a) $a = 13, n = 20$
- (b) $a = 69, n = 89$
- (c) $a = 1891, n = 3797$
- (d) $a = 6003722857, n = 77695236973$

Solution. Use the Euclidean Algorithm to find $x, y \in \mathbb{Z}$ such that $ax + ny = 1$ for each pair (a, n) . Then $ax \equiv 1 \pmod{n}$, and x is the multiplicative inverse of $\bar{a}$ in $\mathbb{Z}/n\mathbb{Z}$.

- (a) 17.
- (b) 40.
- (c) 253.
- (d) 77695236753.

Exercise 0.3.16

Write a computer program to add and multiply mod n , for any n given as input. The output of these operations should be the least residues of the sums and products of two integers. Also include the feature that if $(a, n) = 1$, an integer c between 1 and $n - 1$ such that $\bar{a} \cdot \bar{c} = \bar{1}$ may be printed on request.

REMARK. Your program should not, of course, simply quote “mod” functions already built into many systems.

Solution. Please see `modular_arithmetic.py` for the solution. ■